

- Maxwell's equations
- Ampere's circuit law
 - Faraday's law
 - Gauss's law for electrical field.
 - " " for magnetic field.
- eqns can be written
- in integral form,
 - in point form (differential form)

Maxwell's equations

- (i) for static field.
- (ii) for time varying fields.
- (iii) for good conductor.
- (iv) for harmonically varying fields.

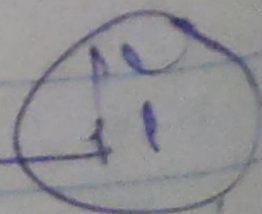
Maxwell's equations for static field.

derived from Faraday's law

$$\oint_L \vec{E} \cdot d\vec{L} = - \int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

$$\oint_C \vec{E} \cdot d\vec{s} = - \int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s}$$

$$(\nabla \times \vec{E}) = - \frac{\partial \vec{B}}{\partial t}$$



(b) derived from Ampere law,

$$\oint \mathbf{H} \cdot d\mathbf{l} = \underline{I}$$

$$I = \int \mathbf{J} \cdot d\mathbf{A}$$

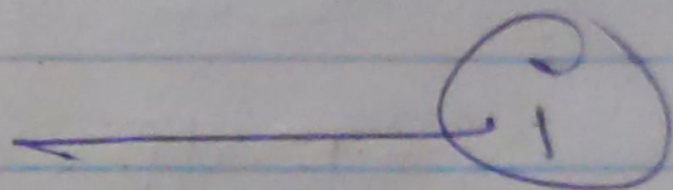
$$\oint d\mathbf{l} = \int \mathbf{J} \cdot d\mathbf{A}$$

$$I = \int \mathbf{J} \cdot d\mathbf{A}$$

$$\oint \mathbf{H} \cdot d\mathbf{l} = \int \mathbf{J} \cdot d\mathbf{A}$$

$$\oint (\nabla \times \mathbf{H}) \cdot d\mathbf{A} = \int \mathbf{J} \cdot d\mathbf{A}$$

$$\nabla \times \mathbf{H} = \mathbf{J}$$



(c) derived from Gauss's law of electric field

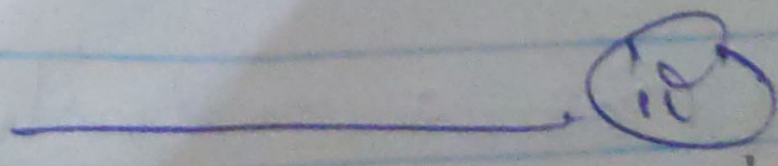
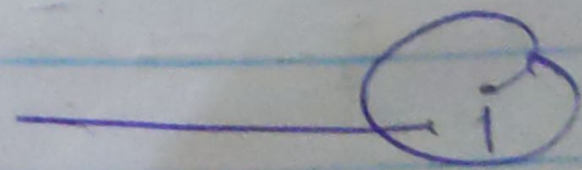
$$\oint \mathbf{D} \cdot d\mathbf{A} = Q_{\text{enclosed}}$$

$$\text{but } Q_{\text{enclosed}} = \int \rho_v \cdot dV$$

$$\oint \mathbf{D} \cdot d\mathbf{A} = \int \rho_v \cdot dV$$

$$\int \nabla \cdot \mathbf{D} \cdot dV = \int \rho_v \cdot dV$$

$$\nabla \cdot \mathbf{D} = \rho_v$$



(4). derived from Gauss's law for magnetic field

$$\oint \vec{B} \cdot d\vec{l} = 0$$

$$\int_V \vec{\nabla} \cdot \vec{B} \, dV = 0$$

$$\vec{\nabla} \cdot \vec{B} = 0$$

* FOR TIME VARYING FIELD.

(5). derived from Faraday's law.

$$\oint \vec{E} \cdot d\vec{l} = - \int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{l}$$

$$\int_V \vec{\nabla} \times \vec{E} \cdot d\vec{l} = - \int \frac{\partial \vec{B}}{\partial t} \cdot d\vec{l}$$

$$\vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

but. $\vec{B} = \mu \vec{H} = \mu_0 \mu_r \vec{H}$

$$\vec{B} = \mu_0 \mu_r \vec{H}$$

$$\vec{\nabla} \times \vec{E} = \frac{\partial}{\partial t} (\mu_0 \mu_r \vec{H})$$

$$\vec{\nabla} \times \vec{E} = - \mu_0 \mu_r \frac{\partial \vec{H}}{\partial t}$$

$$\vec{\nabla} \times \vec{E} = - \mu_0 \mu_r \frac{\partial \vec{H}}{\partial t}$$

(b) derived from Ampere's Circuital law,

$$\oint \mathbf{H} \cdot d\mathbf{l} = I$$

$$\text{but } I = I_c + I_D \Rightarrow \oint \mathbf{H} \cdot d\mathbf{l} = \int \mathbf{J} \cdot d\mathbf{l}$$

$$I = \int \mathbf{J} \cdot d\mathbf{l}$$

$$\oint \mathbf{H} \cdot d\mathbf{l} = \int \mathbf{J} \cdot d\mathbf{l}$$

$$\oint \mathbf{H} \cdot d\mathbf{l} = \int \mathbf{J} \cdot d\mathbf{l}$$

$$I = \int (\mathbf{J}_c + \mathbf{J}_D) \cdot d\mathbf{l}$$

$$\oint \mathbf{H} \cdot d\mathbf{l} = \int (\mathbf{J}_c + \mathbf{J}_D) \cdot d\mathbf{l}$$

$$\oint \mathbf{H} \cdot d\mathbf{l} = \int (\mathbf{J}_c + \mathbf{J}_D) \cdot d\mathbf{l}$$

$$\mathbf{J}_c = \sigma \mathbf{E}$$

$$\mathbf{J}_D = \frac{\partial \mathbf{D}}{\partial t}$$

$$\oint \mathbf{H} \cdot d\mathbf{l} = \int \left(\sigma \mathbf{E} + \frac{\partial \mathbf{D}}{\partial t} \right) \cdot d\mathbf{l} \quad \text{--- (i)}$$

$$\mathbf{H} = \mathbf{J}_c + \mathbf{J}_D \quad \text{--- (ii)}$$

$$\mathbf{H} = \sigma \mathbf{E} + \frac{\partial \mathbf{D}}{\partial t}$$

for harmonic.

$$\nabla \times \vec{H} = \vec{J}_c + \frac{\partial \vec{D}}{\partial t}$$

but

$$\vec{D} = \epsilon \cdot \vec{E}$$

$$\vec{E} = e^{j\omega t}$$

$$\frac{\partial \vec{E}}{\partial t} = j\omega e^{j\omega t}$$

$$\nabla \times \vec{H} = \vec{J}_c + \epsilon \cdot j\omega e^{j\omega t}$$

$$\nabla \times \vec{H} = \vec{J}_c + j\omega \epsilon \vec{E} \quad \text{--- (1)}$$

(c) Gauss's for electric field.

$$\oint \vec{D} \cdot d\vec{r} = Q_{\text{enclosed}}$$

$$Q_{\text{enclosed}} = \int \rho_v \cdot dV$$

$$\oint \vec{D} \cdot d\vec{r} = \int \rho_v \cdot dV$$

$$\int \nabla \cdot \vec{D} \cdot dV = \int \rho_v \cdot dV \quad \text{--- (2)}$$

$$\nabla \cdot \vec{D} = \rho_v \quad \text{--- (3)}$$

(d) Gauss's law for magnetic field.

$$\oint \mathbf{B} \cdot d\mathbf{l} = 0.$$

$$\int \nabla \cdot \mathbf{B} \, dV = 0$$

$$\nabla \cdot \mathbf{B} = 0$$

* Maxwell's eqn for free space:

$$\rho_v = 0, \quad \sigma = 0$$

(a) derived from Faraday's law.

$$\oint (\nabla \times \mathbf{E}) \cdot d\mathbf{s} = - \int \frac{\partial \mathbf{B}}{\partial t} \cdot d\mathbf{s}$$

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}$$

(b) Ampere circuit law,

$$\oint (\nabla \times \mathbf{H}) \cdot d\mathbf{s} = \int \frac{\partial \mathbf{D}}{\partial t} \cdot d\mathbf{s}$$

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t}$$

(c) Gauss's law for electric field.

$$\int \nabla \cdot \mathbf{D} \, dV = 0$$

(d) Gauss's law for magnetic field.

$$\int \nabla \cdot \mathbf{B} \, dV = 0$$

* Maxwell's eqn for good conductor.

$$J = J_c + J_D.$$

$$J = J_c + \frac{\partial D}{\partial t}.$$

$$J_c \gg \frac{\partial D}{\partial t}.$$

$$J = J_c$$

harmonically time

* Maxwell's eqn for good conductor
varying field:

$$\vec{D} = D_0 e^{i\omega t}$$

$$\vec{B} = B_0 e^{i\omega t}$$

$$\frac{\partial \vec{D}}{\partial t} = i\omega D_0 e^{i\omega t}$$

$$\frac{\partial \vec{B}}{\partial t} = i\omega B_0 e^{i\omega t}$$

$$\frac{\partial \vec{D}}{\partial t} = i\omega [\vec{D}_0 e^{i\omega t}]$$

$$\frac{\partial \vec{D}}{\partial t} = i\omega \vec{D}$$

$$\frac{\partial \vec{B}}{\partial t} = i\omega \vec{B}$$

$$H = H_m \cos \beta_2 \cos \omega t \hat{a}_y \text{ A/m}$$

$\vec{E}$, $\vec{D}$ and $\vec{B}$.

$$\text{For } \nabla \times \vec{E} = - \frac{\partial \vec{B}}{\partial t}$$

$$\nabla \times \vec{E} = - \frac{\partial}{\partial t} (\mu \vec{H}) = -\mu_0 \mu_r \frac{\partial \vec{H}}{\partial t}$$

$$\nabla \times \vec{E} = -\mu_0 \frac{\partial \vec{H}}{\partial t} \quad \text{--- (1)}$$

$$\vec{B} = \mu_0 \vec{H}$$

$$\vec{B} = 4\pi \times 10^{-7} [H_m \cos \beta_2 \cos \omega t] \hat{a}_y \text{ T}$$

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$$

$$\nabla \times \vec{H} = \begin{vmatrix} \hat{a}_x & \hat{a}_y & \hat{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & H_m \cos \beta_2 \cos \omega t & 0 \end{vmatrix}$$

$$\nabla \times \vec{H} = - \frac{\partial}{\partial z} (H_m \cos \beta_2 \cos \omega t) \hat{a}_x + 0 \hat{a}_y +$$

$$\frac{\partial}{\partial x} (H_m \cos \beta_2 \cos \omega t)$$

$$\vec{v} \times \vec{H} = \mu_m B_1 \sin \beta_2 \cos \omega t \quad \text{--- (1)}$$

$$\vec{v} \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$$

$$\vec{v} \times \vec{H} = \epsilon \frac{\partial \vec{E}}{\partial t}$$

$$\mu_m B_1 \sin \beta_2 \cos \omega t = \frac{\partial \vec{D}}{\partial t}$$

$$\int \mu_m B_1 \sin \beta_2 \cos \omega t dt = \vec{D}$$

$$\vec{D} = \frac{\mu_m B_1 \sin \beta_2 \sin \omega t}{\omega} \quad \text{C/m}^2$$

$$\vec{E} = \left(\frac{\mu_m B_1 \sin \beta_2 \sin \omega t}{\omega \epsilon} \right) \text{ N/C }$$